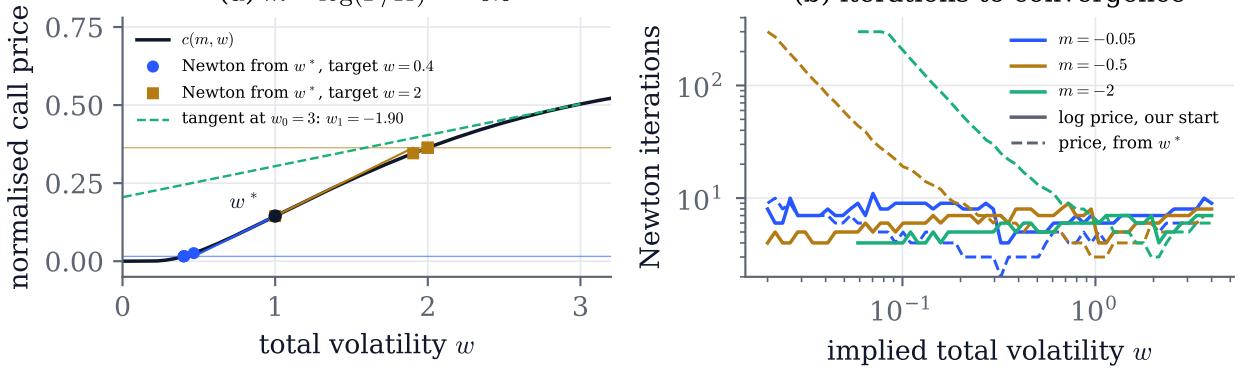
(a) $m = \log(F/K) = -0.5$
normalised call price
0.75
0.50
0.25
0.00
0
1
2
3
total volatility w
$c(m, w)$
Newton from w^*, target $w = 0.4$
Newton from w^*, target $w = 2$
tangent at $w_0 = 3$: $w_1 = -1.90$
w^*
(b) iterations to convergence
Newton iterations
10^2
10^1
10^{-1}
10^0
implied total volatility w
$m = -0.05$
$m = -0.5$
$m = -2$
log price, our start
price, from w^*